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Statistical Methods

Statistical analysis was done using R software version 4.5.1. **Descriptive statistics** for the Pre-manifest HD Carriers and the Controls were described as counts and percentages for each, with the *group* variable. Mean, standard deviation, median and range were used for age (years), Symbol Digit Modalities test (SDMT) score and CAG repeat length, overall, and by group. Box plots, Q-Q plots and histograms were used to investigate distribution and difference of age, SDMT and CAG within the groups. Pearson correlation coefficients and scatter plots were employed to presence and strength of the linear relationships between SDMT, age and CAG repeat length. Correlations tests were used to assess significance of the relationships.

Simple linear regression models were used to compare difference in average SDMT scores (the continuous outcome) and group (binary predictor categorised as Pre-manifest HD Gene Carriers and Controls). Adjusted linear regression models were used to check presence of any significant differences in average SDMT score within three levels of a subgroup variable "*sdmt_cat*" (level 1 – controls, level 2 – HD gene carriers with CAG repeat length ≤ 42 and level 3 – HD Gene carriers with CAG repeat length ≥ 43) with age included as a confounder.

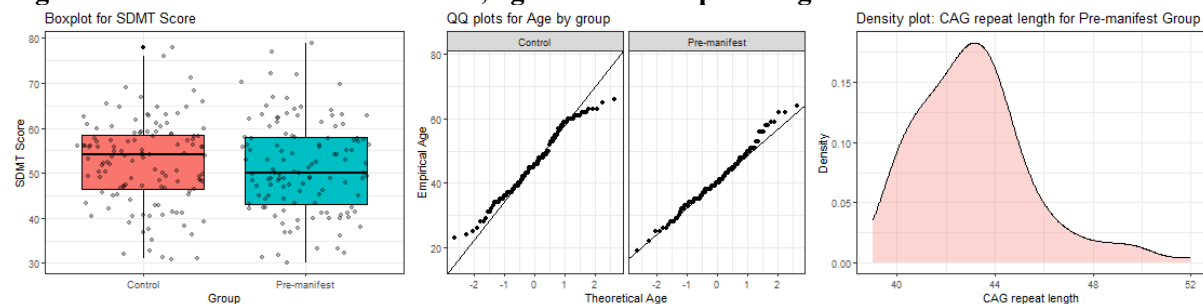
Plots for residuals of adjusted models were used to check for their robustness. QQ plots, residuals vs fitted values and residuals vs age plots were used to assess normality, homoscedasticity, and linear relationship of the residuals with **the continuous age predictor**. An interaction model was also run to assess the assumption of no interaction within categories.

Results

Exploratory analysis

The SDMT scores for both groups were approximately normally distributed (Figure 1), with a slightly higher median in the control group. Visual exploration of the SDMT between the two groups suggests a minimal right skew for both groups and slight heavy tails indicating a modest number of observations at the extremes. Age is slightly skewed within the controls and Pre-manifest HD Gene carriers. CAG repeat lengths for the Pre-manifest HD gene carriers were right-skewed, with more observations having a higher value (Figure 1).

Figure 1: Distribution of SDMT score, age and CAG repeat length



The proportions of participants in the control group (50.6%, n=123) and the pre-manifest HD Gene carrier group (49.4%, n=120) were slightly balanced. However, average age and SDMT score were higher in the controls (mean age 46.195 years, mean SDMT score 52.171 units) as compared to the Pre-manifest HD Gene carriers (mean age 40.817 years, mean SDMT score 51.208 units) (Table 1).

Table 1: Summary statistics for Pre-manifest HD Gene Carriers and Controls

Variable	Statistic	Overall	Controls	Pre-manifest HD Gene Carriers
N(%)	Count (%)	243(100)	123(50.617)	120(49.383)
Age (years)	Mean (SD)	43.539(9.945)	46.195(10.261)	40.817(8.856)
	Median (Range)	43(19-66)	46(23-66)	40(19-64)
SDMT Score	Mean (SD)	51.695(9.809)	52.171(9.462)	51.208(10.169)
	Median (Range)	52(30-79)	54(31-78)	50(30-79)
CAG repeat length	Mean (SD)	-	-	43.142(2.398)
	Median (Range)	-	-	43(39-52)

Pearson correlation analysis showed a weak but significant negative linear relationship between SDMT score and age ($r=-0.2403$, $p\text{-value}=0.0001$, CI: -0.355, -0.118). However, SDMT score and CAG repeat length had an extremely weak insignificant relationship ($r=-0.0002$, $p\text{-value}=0.998$). Age and CAG repeat length showed a very strong negative linear relationship ($r=-0.819$, $p\text{-value}<0.0001$, CI: -0.8709, -0.7507).

Comparison of SDMT scores between groups

Table 2 below shows the crude and adjusted estimates for the average SDMT score of the controls and Pre-HD manifest carriers.

Table 2: Crude and adjusted results for SDMT regression model: Group and age -adjusted

	Crude Slope (95% CI)	p-value	Adjusted slope (95% CI)	p-value
Intercept	52.171 (50.427, 53.915)	<0.0001	52.888 (51.170, 54.606)	<0.0001
Pre-manifest HD Gene carriers	-0.9624 (-3.444, 1.519)	0.446	-2.415 (-4.906, 0.077)	0.058
Age*			-0.270 (-0.396, -0.144)	<0.0001

*Age centered at its mean

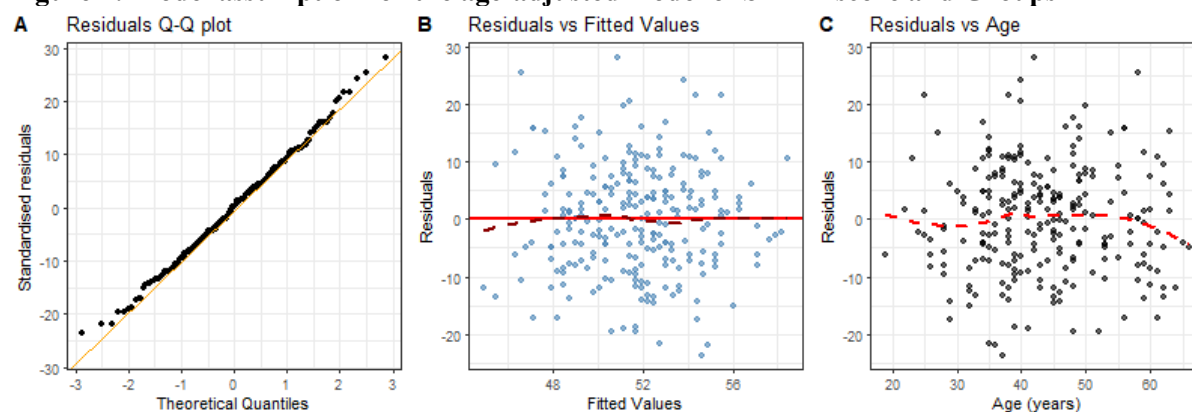
The crude simple linear regression model showed a lower non-significant average SDMT score for the Pre-manifest HD Gene carriers compared to the controls (Slope = -0.9624; p-value=0.446). The average SDMT score for controls was 52.171 units (CI: 50.427, 52.915; p-value<0.0001).

In the age-adjusted model (Table 2), the average SDMT score for the controls at the mean was 52.888 (95%CI: 51.170, 54.606; p-value<0.0001). Adjusting for age led to a **more negative difference** in average SDMT score for Pre-manifest HD carriers (Adj slope: -2.415; 95% CI: -4.906, 0.077) for participants with the same mean age. The difference, however, is borderline statistically significant (p-value=0.058). The model also showed that for an increase in age by a year, the average SDMT score for both groups reduced by 0.27 units (95% CI: -0.396, -0.144 and p-value<0.0001).

The exploratory analysis and crude model show a similar difference in average SDMT (approx. 0.96), but there is no clear evidence of its existence (p-value=0.446). Adjusting for age, however, results in a **more negative difference** in average SDMT between the groups, with slightly clearer evidence (p-value=0.058). The adjusted model therefore suggests that the small difference in average SDMT for the two groups is potentially masked by the age effects in the crude model, since the controls were older than the controls on average, and age had a significant negative relationship with SDMT ($r=-0.2403$).

The adjusted model was robust meeting the assumptions for normality, homoscedasticity and linearity of the residuals (Figure 2).

Figure 2: Model assumption for the age-adjusted model of SDMT score and Groups



Comparison of SDMT scores between subgroups

Table 3 below shows that 50.62% (n=123) of the participants are healthy controls (Category 1), and the HD Gene Carriers distributed as follows: 49(20.16%) with CAG repeat length less or equal to forty-two (Category 2), and 71(29.22%) with CAG repeat length greater or equal to forty-three (Category 3).

The category 2 group were the oldest (mean age 48.184 years), followed by the controls (mean age 46.195 years) while those with the higher CAG repeat length (≥ 43) were the youngest (mean age 36 years). Regarding the cognitive score, the controls had the highest SDMT score (mean 52.171, median 54), while the category 3 (mean 51.352) participants had a slightly higher SDMT score than the category 2 (mean 51.000). As expected, the CAG repeat lengths were significant different between the HG Gen carrier groups, with category 2 averaging 41.020 repeats (SD=0.901) and category 3 averaging 44.606 repeats (SD=1.975).

Table 3: Summary statistics for Pre-manifest HD Gene Carrier categories and Controls

Variable	Statistic	Overall	Controls (1)	HD Gene Carriers with CAG≤42 (2)	HD Gene Carriers with CAG≥43 (3)
N(%)	Count (%)	243(100)	123(50.62)	49(20.16)	71(29.22)
Age (years)	Mean (SD)	43.539(9.945)	46.195(10.261)	48.184(7.076)	35.732(5.899)
	Median(Range)	43(19-66)	46(23-66)	48(34-64)	36(19-50)
SDMT Score	Mean (SD)	51.695(9.809)	52.171(9.462)	51.000(9.806)	51.352(10.479)
	Median(Range)	52(30-79)	54(31-78)	52(30-72)	50(31-79)
CAG repeat length	Mean (SD)	-	-	41.020(0.901)	44.606(1.975)

The crude regression model for average SDMT for the subgroups shows **no statistically significant** difference in SDMT score for either category 2 (Slope = -1.171; 95% CI: -4.444, 2.103; p-value=0.482) or category 3 (Slope = -0.819; 95% CI: -3.707, 2.070; p-value=0.577) compared to the controls. However, on adjusting for age, **a strong statistically significant** effect of age is identified, showing that SDMT score decreased with a year increase in age by 0.331 units (95% CI: -0.471, -0.192; p-value<0.0001). Additionally, the category 2 (Adj slope = -0.512; 95% CI: -3.664, 2.641; p-value=0.749) **difference remains non-significant**, but the category 3 had a significantly lower SDMT score of -4.286 (95% CI: -7.419, -1.153; p-value=0.008) compared to the controls at the **average age**.

Table 4: Crude and adjusted results for SDMT regression model: Group and age -adjusted

	Crude Slope (95% CI)	p-value	Adjusted slope (95% CI)	p-value
Group				
Intercept	52.171 (50.423, 53.918)	<0.0001	53.051 (51.334, 54.768)	<0.0001
HD Gene Carriers with CAG≤42	-1.171 (-4.444, 2.103)	0.482	-0.512 (-3.664, 2.641)	0.749
HD Gene Carriers with CAG≥43	-0.819 (-3.707, 2.070)	0.577	-4.286 (-7.419, -1.153)	0.008
Age*			-0.331 (-0.471, -0.192)	<0.0001

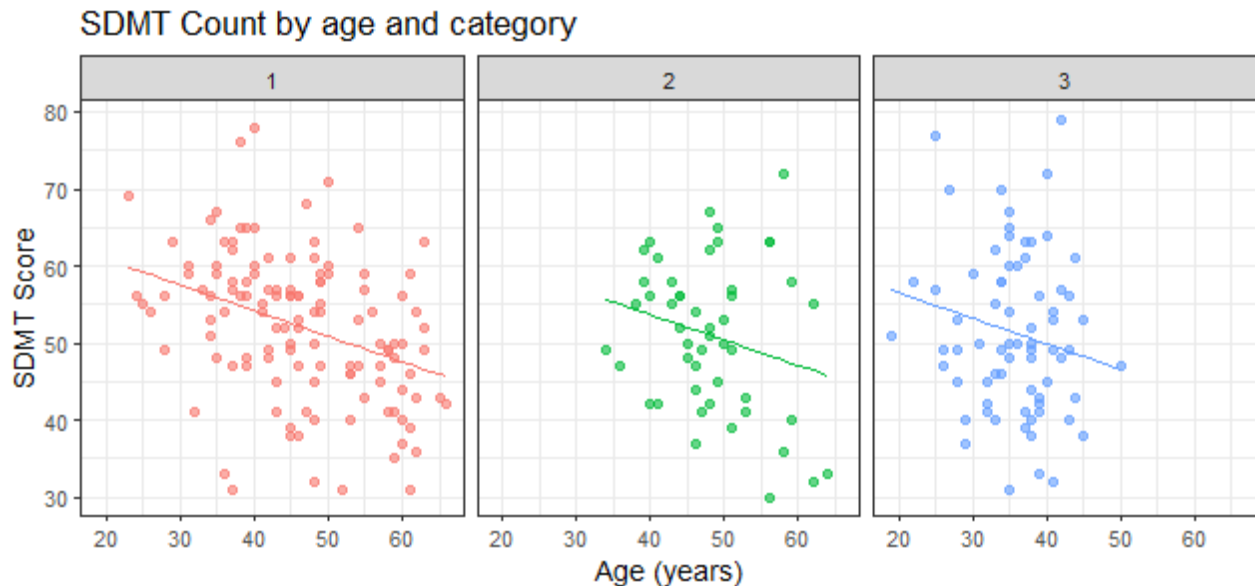
*Age centered at its mean

From question 1 and question 3a, the older participants are those in the control group. Additionally, age and CAG realise a very strong negative linear relationship. In the unadjusted model we see that the differences in average SDMT score between the HD Gene carriers and controls are insignificant. However, in the age-adjusted model, we note a highly significant and **more negative effect** of age on the difference of HD Gene carriers with CAG ≥43 compared to the controls. **We also note that the effect of age on the average SDMT is highly significant, indicating that age is a very strong confounder for this younger group.**

For the age-adjusted model, normality of the SDMT within the three groups is met. Additionally, the assumptions of uniform variance residuals, and linearity of residuals are equally met. **Lastly, a Q-Q plot revealed that the residuals for the age-adjusted model were very slightly positively skewed.**

To assess the assumption of no-interaction a linear regression model with the interaction term (age*sdmt_cat) was run, and this showed insignificant slope values at 95% confidence level, indicating that the assumption was met. This assumption is further explained by figure 3 below.

Figure 3: Regression lines plot of predicted values of average SDMT against age within groups



Discussion

From the exploratory analysis of our data, it is seen that participants without the HD Gene were older and had a higher cognitive performance than those with the Pre-manifest HD Gene, but this was not significant. However, we notice that without consideration of the CAG repeat length, the controls also turned out to have a higher cognitive performance than the Pre-manifest gene carriers when the effect of age was considered. We did not have evidence to show this difference. On subgrouping the Pre-manifest HD gene carriers into those with $CAG \leq 42$ and ≥ 43 , we notice that participants with higher CAGs were significantly younger than all the other participants. On testing the association of these with SDMT cat we had clear evidence that age increased the negative difference between young people with high CAGs and the older controls, underpinning the relevance and confounding effect of age in cognitive function of individuals with high CAG.

From this study, we do not have the CAG repeat lengths measured for the controls and yet they have an average SDMT score, which is almost half the max possible score. While the gene carriers have lower SDMT scores than the controls, the values for the controls do not warrant complete immunity from HD.

Additionally, the study only collected data on age as a confounder. The true difference in cognitive function between controls and Pre-manifest HD carriers could be explained by any other factors apart from age alone. It would have been ideal to have data on more factors like sex, genetics and lifestyle assessments, to make robust conclusions on difference between the groups.

While I believe the estimates for the preferred age-adjusted regression model of subcategories and is robust for this analysis, the assumption of linearity between age and SDMT may not be met fully met (non-linear relationships between age and the fitted values especially at the extremes). This shows a potential non-linear relationship between age and SDMT and we could have dealt with this by using splines for age to interpret the slopes for the different groups appropriately.

Appendix

Age-adjusted model in On 3b

Let Y_i be a continuous dependent variable with a X_i dependent variable for the i th participant. Then, for a linear regression model; $y_i = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$,

$$\text{where } \epsilon_i \sim NID(0, \sigma^2)$$

The values of i are the observations in our dataset, i.e. $i = 1, 2, \dots, 243$

If X_1 is a categorical variable with three levels 1, 2, 3, then;

$$u_{1i} \begin{cases} 1 & \text{if } X_1 = 2 \\ 0 & \text{if } X_1 \neq 2 \end{cases}$$

$$u_{2i} \begin{cases} 1 & \text{if } X_1 = 3 \\ 0 & \text{if } X_1 \neq 3 \end{cases}$$

Therefore, our equation would be interpreted as:

$$y_i = \alpha + \delta_1 u_{1i} + \delta_2 u_{2i} + \beta_2 X_{2i} + \epsilon_i$$

Putting it in the context of the regression equation in 3b, we get:

$$SDMT_i = 53.051 - 0.512(HD \text{ CAG} \leq 42)_i - 4.286(HD \text{ CAG} \geq 43)_i - 0.331Age_{centered,i} + \epsilon_i$$

With:

- α being the expected value of the SDMT for the controls at the mean age.
- δ_1 is the difference between the average SDMT of the controls and the individuals with HD Gene carriers with CAG repeat length ≤ 42 , while δ_2 is the difference between average SDMT for the controls and the HD Gene carriers with CAG repeat length ≥ 43 . For both coefficients, age is held constant.
- β_2 is the change in average SDMT for an increase in age by a year, for all the groups. This slope is assumed to be the same for all the levels of the categorical variable.
- ϵ_i is the error term, and is defined by the random error for the i 'th participant
- σ is the standard deviation of the residuals for the model

The table below therefore shows the effective regression equations for each individual group:

Parameter	Regression equation
Healthy Controls	$y_i = \alpha + \beta_2 Age_i + \epsilon_i$
$CAG \leq 42$	$y_i = \alpha + \delta_1(HD \text{ CAG} \leq 42)_i + \beta_2 Age_i + \epsilon_i$
$CAG \geq 43$	$y_i = \alpha + \delta_2(HD \text{ CAG} \geq 43)_i + \beta_2 Age_i + \epsilon_i$

For the extended model of interaction:

We have: $y_i = \alpha + \beta_1 HD \text{ CAG} \leq 42_i + \beta_2 HD \text{ CAG} \geq 43_i + \beta_3 Age_{centered,i} + \beta_4 (Age_{centered,i} * HD \text{ CAG} \leq 42_i) + \beta_5 (Age_{centered,i} * HD \text{ CAG} \geq 43_i) + \epsilon_i$

$$\text{where } \epsilon_i \sim NID(0, \sigma^2)$$

Where:

- β_3 is the change in SDMT score an increase in age by a year, specific to the controls
- β_4 is the difference in age effect on SDMT score between $CAG \leq 42$ and the controls
- β_5 is the difference in age effect on SDMT score for $CAG \geq 43$ and the controls
- $\alpha + \beta_1$ is mean SDMT score for $CAG \leq 42$ at the mean age.
- $\alpha + \beta_2$ is mean SDMT score for $CAG \geq 43$ at the mean age.
- $\beta_3 + \beta_4$ is the change in SDMT score for $CAG \leq 42$ group. This basically is the difference combined with the additional effect of being in the $CAG \leq 42$ group.
- $\beta_3 + \beta_5$ is the change in SDMT score for the $CAG \geq 43$ group